

Cyber Security Internship Report

Week 5 – Task 15 (SOC)

Incident Detection – Investigate an Attack in Logs

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TABLE OF CONTENTS

1. INTRODUCTION.....	3
2. TASK OBJECTIVE.....	3
3. LAB ENVIRONMENT.....	3
4. INVESTIGATION METHODOLOGY.....	3
5. WINDOWS EVENT ID 4625 INVESTIGATION.....	4
6. DETAILED EVENT ANALYSIS.....	5
7. FAILED LOGIN ATTEMPT ANALYSIS.....	6
8. SUCCESSFUL LOGON INVESTIGATION.....	7
9. INCIDENT TIMELINE.....	9
10. INCIDENT FINDINGS.....	9
11. TRYHACKME PRACTICAL TRAINING.....	10
12. SKILLS PRACTICED.....	11
13. SECURITY SIGNIFICANCE.....	12
14. CONCLUSION.....	12
15. EVIDENCE / SCREENSHOT INDEX.....	12

1. INTRODUCTION

Authentication events in the Windows Security log are a primary data source for detecting suspicious access activity. Repeated failed logons (Event ID 4625) from a single source, and successful logons (Event ID 4624) from the same source, may indicate that credentials are being guessed or misused, or they may have a legitimate explanation. Log analysis is needed to establish which.

This report documents Week 5 – Task 15 (SOC). The Security log of a Windows Server 2022 virtual machine was investigated using Windows Event Viewer and PowerShell, and the TryHackMe rooms Investigating Windows and Juicy Details were completed. The report keeps observed evidence separate from interpretation, and states where the provided screenshots do not demonstrate a detail.

2. TASK OBJECTIVE

The practical investigation was intended to identify the following:

- Failed authentication activity in the Windows Security log.
- The source IP address associated with the failed activity.
- The failed attempt count for that source IP address.
- Successful logons from the same source IP address.
- An attack/investigation timeline based on the observed timestamps.
- Practical log investigation skills using Event Viewer and PowerShell.

The task also required completing the TryHackMe rooms Investigating Windows and Juicy Details.

3. LAB ENVIRONMENT

Only environments and tools shown in the screenshots are listed.

Item	Detail shown in the evidence
Operating system	Windows Server 2022, running as a VirtualBox virtual machine (window title: WinServer2022-Agent [Running] - Oracle VirtualBox)
Computer name	WIN-BCDCP359AAS (shown in the Event Viewer event details)
Log source	Windows Security Event Log, Number of events: 26,838
Tools	Windows Event Viewer; Windows PowerShell (Administrator: Windows PowerShell, prompt PS C:\Users\Administrator>)
Training platform	TryHackMe, accessed in Firefox inside a Kali Linux virtual machine (window title visible in Figure 6)
Screenshot dates	Windows screenshots show the system clock 20-09-2026, between 03:34 PM and 03:46 PM

No SIEM platform or additional analysis tool is shown in the evidence. Timestamps appear in the screenshots in day-month-year format with dot separators (for example 11.53.21 AM); they are written with colons in this report.

4. INVESTIGATION METHODOLOGY

The steps below follow the order of the investigation. The final column identifies the screenshot that supports each step.

Step	Action	Supporting evidence
1	Open Windows Event Viewer.	Figure 1 (Event Viewer window)
2	Navigate to Windows Logs > Security.	Figure 1 (Security selected under Windows Logs)

Step	Action	Supporting evidence
3	Filter the Security log for Event ID 4625.	Figure 1 (filter line: Log: Security; Event ID: 4625; 296 events). The filter dialog itself is not captured.
4	Inspect a failed logon event.	Figure 1 (General tab of the selected event)
5	Extract the source IP address and related authentication information.	Figure 2 (EventData fields)
6	Use PowerShell to investigate failed logons associated with the source IP.	Figure 3
7	Check Event ID 4624 for successful logons from the same source IP.	Figures 4 and 5
8	Build a timeline from the observed timestamps.	Section 9 (based on Figures 1, 4, and 5)
9	Complete the required TryHackMe rooms.	Figures 6 and 7

5. WINDOWS EVENT ID 4625 INVESTIGATION

Figure 1 shows the Security log in Event Viewer with a filter applied for Event ID 4625, which represents a failed logon attempt. The filter line reads “Filtered: Log: Security; Source: ; Event ID: 4625. Number of events: 296”, while the unfiltered Security log contains 26,838 events. The values below are taken from the selected event in Figure 1 and from the event details in Figure 2.

Item	Value shown	Source
Event ID	4625	Figure 1
Event type (Keywords)	Audit Failure	Figure 1
Event description	An account failed to log on.	Figure 1
Level / Task Category	Information / Logon	Figure 1
Event time	15-09-2026 11:53:21 AM	Figure 1
Target account	Administrator	Figure 2
Source IP	10.209.222.48	Figure 2
Logon Type	3	Figure 2
Authentication Package	NTLM	Figure 2
Source Port (lpPort)	58690	Figure 2

Observations from the event list. The visible rows are all Audit Failure entries with Event ID 4625 and Task Category Logon. The first row is 15-09-2026 11:53:21 AM, followed by 11:53:06 AM, 11:53:04 AM, and 11:26:30 AM. A run of consecutive entries stamped 10:56:47 AM fills the remainder of the visible list. The source IP of each individual entry is not shown in the list view.

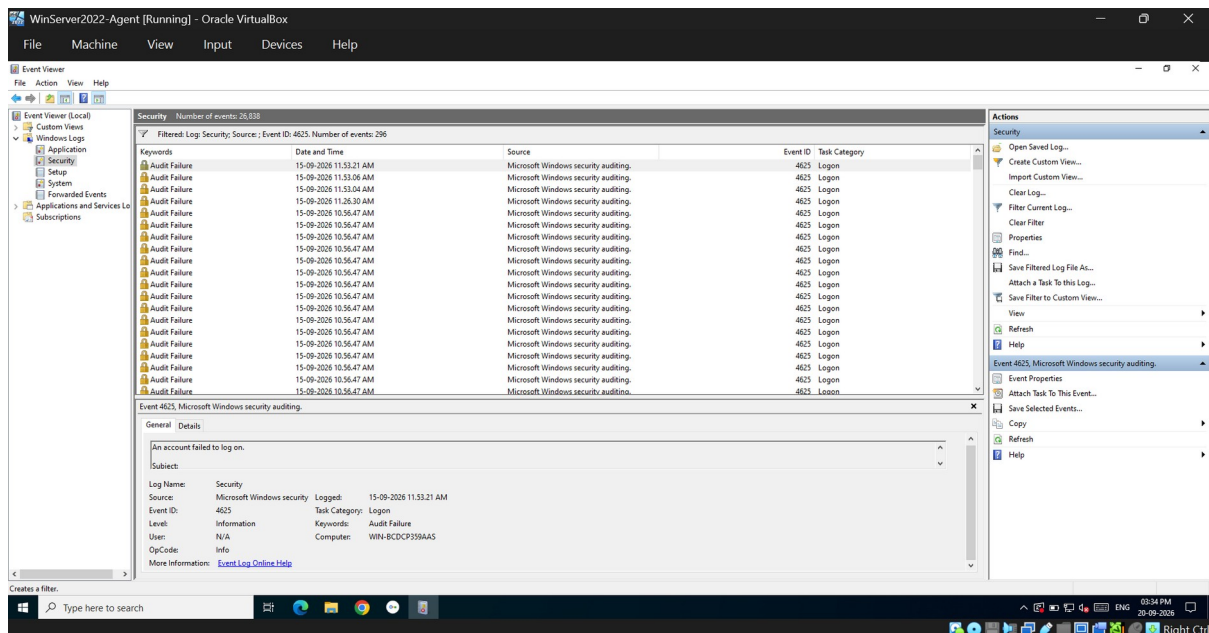


Figure 1 - 01_security_log_event_4625_failed_logins.png
Windows Security Event Viewer showing filtered Event ID 4625 failed logon events.

6. DETAILED EVENT ANALYSIS

Figure 2 shows the Details tab of the Event Properties window for an Event ID 4625 event, in Friendly View. The System section is collapsed, so the event timestamp is not visible in this dialog. The association of these details with the 11:53:21 AM entry in Figure 1 is taken from the task information and is not independently shown in the screenshot.

EventData field	Value shown	Significance
TargetUserName	Administrator	The account against which the logon attempt was made.
TargetDomainName	WIN-BCDCP359AAS	Same as the computer name shown in Figure 1, so the attempt referenced an account on this machine.
IpAddress	10.209.222.48	The source address of the failed logon; used as the pivot for the rest of the investigation. It is a private-range address, and the identity of the host is not demonstrated in the evidence.
IpPort	58690	The port on the source host from which the connection originated.
LogonType	3	Logon type 3 is a network logon in Windows documentation.
LogonProcessName / AuthenticationPackageName	NtLmSsp / NTLM	The failed attempt used NTLM authentication.
Status / SubStatus	0xc000006d / 0xc000006a	Documented for Event 4625 as a generic logon failure, with the sub-status commonly indicating an incorrect password. The status text is not displayed in the screenshot.
FailureReason	%%2313	A message resource identifier. Its text is not displayed in the screenshot.

EventData field	Value shown	Significance
SubjectUserSid / SubjectUserName / SubjectDomainName / SubjectLogonId	S-1-0-0 / - / - / 0x0	No subject account is identified for this event.
WorkstationName, TransmittedServices, LmPackageName, ProcessName	- (not populated)	No value recorded for these fields in this event.
KeyLength / ProcessId	0 / 0x0	No value recorded beyond the defaults shown.

The fields show a failed network logon using NTLM against the Administrator account from 10.209.222.48. They do not, on their own, show the intent behind the attempt or how many attempts were made; the count is examined in Section 7.

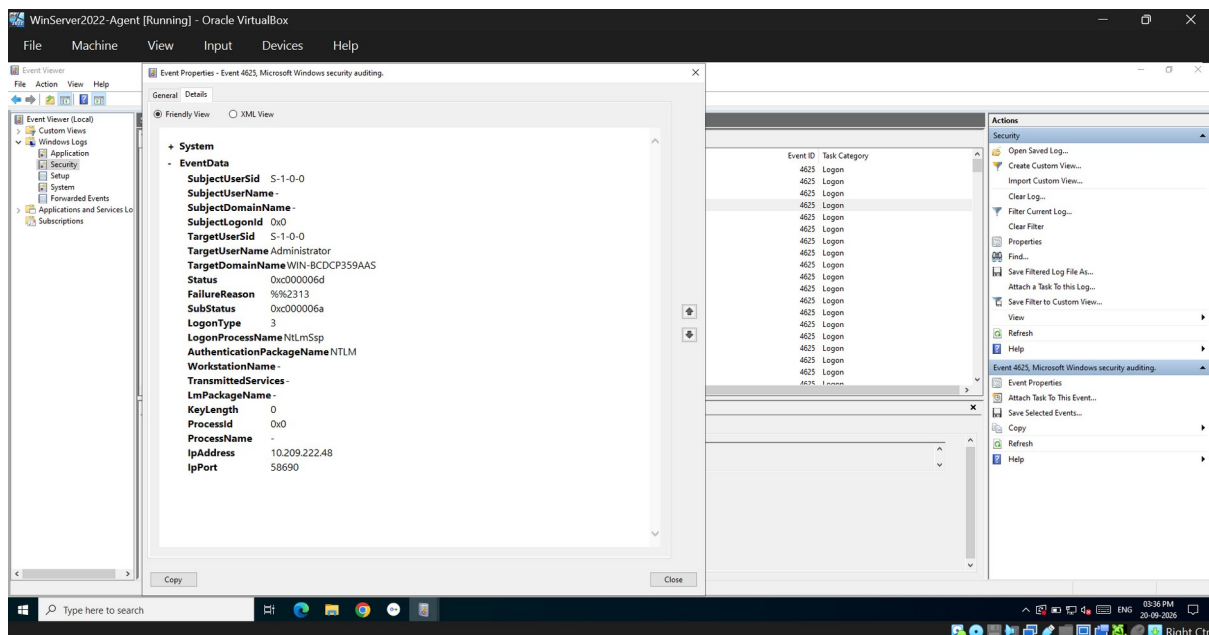


Figure 2 – 02_event_4625_source_ip_administrator.png
Event ID 4625 properties showing the target account and source IP address.

7. FAILED LOGIN ATTEMPT ANALYSIS

PowerShell was used to identify Event ID 4625 events associated with source IP 10.209.222.48. The command shown in Figure 3 retrieves the Security log events with Id 4625, converts each event to XML, reads the IpAddress data element, outputs the event when it equals 10.209.222.48, and counts the output.

```
$events = Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625}
$events | ForEach-Object {
    $xml = [xml]$_ .ToXml()
    $ip = ($xml.Event.EventData.Data | Where-Object {$_.Name -eq
    'IpAddress'}).'#text'
    if ($ip -eq '10.209.222.48') { $_ }
} | Measure-Object | Select-Object Count
```

Result. The output shows Count 43. This is the count returned by the query shown in the evidence.

XML parsing error. The output also contains a red error for one event: the event XML could not be converted to type System.Xml.XmlDocument, with the message that a hexadecimal value 0x1B is an invalid character at line 1, position 882 (FullyQualifiedErrorId: InvalidCastToXmlDocument). The XML

printed in the error belongs to a different event (EventRecordID 7314, SystemTime 2026-06-02T07:35:11Z) whose IpAddress is 172.29.57.48, with WorkstationName kali and TargetUserName Administrator. That event would therefore not have matched 10.209.222.48 even if it had been parsed. Only one such error is visible in the captured output, but the screenshot does not show whether any other event was skipped.

Interpretation. Because the command reported a parsing error, 43 should be read as the count produced by this query, not as a fully validated count of every failed logon from 10.209.222.48. The source IPs of the remaining Event ID 4625 events in the 296-event filtered view are not demonstrated in the evidence.

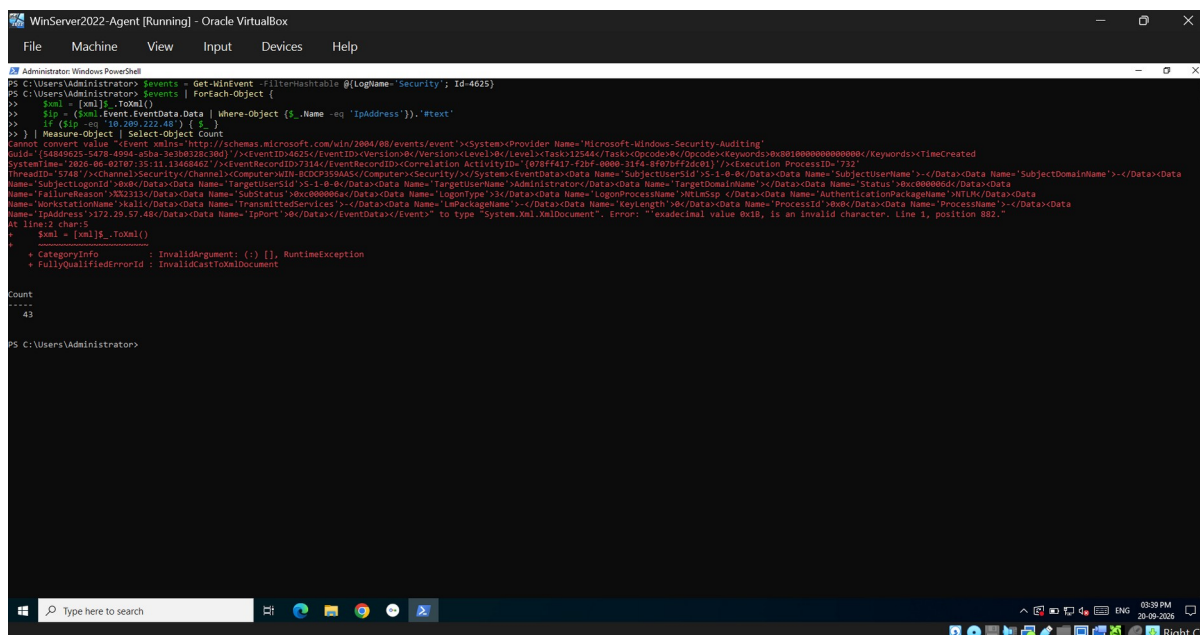


Figure 3 – 03_powershell_43_failed_login_attempts.png

PowerShell query showing the returned count of failed logon events associated with the investigated source IP.

8. SUCCESSFUL LOGON INVESTIGATION

Event ID 4624 represents a successful logon. Two PowerShell queries were run against the Security log for source IP 10.209.222.48.

Query 1 (Figure 4). Events with Id 4624 were filtered by matching the IP address in the message text, then displayed with TimeCreated, Id, and Message:

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4624} |
Where-Object { $_.Message -match '10\209\222\48' } |
Select-Object TimeCreated, Id, Message
```

The output lists 12 records, each with Id 4624 and the message beginning “An account was successfully logged on”. The message text is truncated in the display, so the account name and logon type of these records are not shown.

```
WinServer2022-Agent [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Administrator: Windows PowerShell
Windows PowerShell
Copyright (c) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-WinEvent -FilterHashtable @(LogName='Security'; Id=4624) |
> Where-Object { $_.Message -match '10.209.222.48' } |
> Select-Object TimeCreated, Id, Message

TimeCreated      Id Message
-----
15-09-2026 1:18:28 PM 4624 An account was successfully logged on...
15-09-2026 12:46:46 PM 4624 An account was successfully logged on...
15-09-2026 11:54:14 AM 4624 An account was successfully logged on...
15-09-2026 11:54:09 AM 4624 An account was successfully logged on...
15-09-2026 11:30:25 AM 4624 An account was successfully logged on...
15-09-2026 11:30:25 AM 4624 An account was successfully logged on...
15-09-2026 11:27:08 AM 4624 An account was successfully logged on...
15-09-2026 11:25:11 AM 4624 An account was successfully logged on...
15-09-2026 10:59:47 AM 4624 An account was successfully logged on...
15-09-2026 10:57:34 AM 4624 An account was successfully logged on...
15-09-2026 10:56:47 AM 4624 An account was successfully logged on...
15-09-2026 10:56:00 AM 4624 An account was successfully logged on...
```

Figure 4 - 04_successful_logon_events_from_source_ip_10.209.222.48.png
PowerShell output showing successful Event ID 4624 logons associated with source IP 10.209.222.48.

Query 2 (Figure 5). The same Id 4624 events were parsed as XML and compared on the IpAddress data element, displaying TimeCreated, EventID, and SourceIP for exact matches with 10.209.222.48. The output again lists 12 records, with the same timestamps as Figure 4 and SourceIP 10.209.222.48 in each row, and no error is visible. The screenshot file is named with “failed_logon_attempt_count”, but the output it contains is a listing of Event ID 4624 records, so it is used here as supporting evidence for the successful logons. The two queries use different matching methods and return the same 12 records.

```
WinServer2022-Agent [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-WinEvent -FilterHashtable @(LogName='Security'; Id=4624) | ForEach-Object {
> $xml = [xml]$_.ToXml()
> $ip = ($xml.Event.EventData.Data | Where-Object {$_.Name -eq 'IpAddress'}).Text
> If ($ip -eq '10.209.222.48') {
> [PSCustomObject]$?
> TimeCreated = $_.TimeCreated
> EventID = $_.Id
> SourceIP = $ip
> }
> }

TimeCreated      EventID SourceIP
-----
15-09-2026 1:18:28 PM 4624 10.209.222.48
15-09-2026 12:46:46 PM 4624 10.209.222.48
15-09-2026 11:54:14 AM 4624 10.209.222.48
15-09-2026 11:54:09 AM 4624 10.209.222.48
15-09-2026 11:30:25 AM 4624 10.209.222.48
15-09-2026 11:30:25 AM 4624 10.209.222.48
15-09-2026 11:27:08 AM 4624 10.209.222.48
15-09-2026 11:25:11 AM 4624 10.209.222.48
15-09-2026 10:59:47 AM 4624 10.209.222.48
15-09-2026 10:57:34 AM 4624 10.209.222.48
15-09-2026 10:56:47 AM 4624 10.209.222.48
15-09-2026 10:56:00 AM 4624 10.209.222.48
```

Figure 5 - 05_failed_logon_attempt_count_from_source_ip_10.209.222.48.png
PowerShell output listing Event ID 4624 records parsed from event XML, with SourceIP 10.209.222.48.

Observation. Successful Event ID 4624 logons from 10.209.222.48 were observed on 15-09-2026 between 10:56:00 AM and 1:18:28 PM. The evidence does not show which accounts these logons used, their logon types, or whether they were authorised, so they are not characterised as attacker access.

9. INCIDENT TIMELINE

The timeline uses only timestamps visible in Figures 1, 4, and 5. All entries are dated 15-09-2026. Shaded rows are failed logon activity (Event ID 4625); unshaded rows are successful logon activity (Event ID 4624).

Time	Event ID	Activity	Source IP	Observation
10:56:00 AM	4624	Successful logon	10.209.222.48	First successful logon record in the queried output.
10:56:47 AM	4625	Failed logon (Audit Failure)	Not shown in list view	Run of consecutive 4625 entries with this same timestamp in Figure 1.
10:56:47 AM	4624	Successful logon	10.209.222.48	Same timestamp as the 4625 run; order within the second is not determinable.
10:57:34 AM	4624	Successful logon	10.209.222.48	Observed about 47 seconds after 10:56:47 AM.
10:59:47 AM	4624	Successful logon	10.209.222.48	Observed.
11:25:11 AM	4624	Successful logon	10.209.222.48	Observed.
11:26:30 AM	4625	Failed logon (Audit Failure)	Not shown in list view	Single 4625 entry at this time in Figure 1.
11:27:08 AM	4624	Successful logon	10.209.222.48	Observed 38 seconds after the 11:26:30 AM failed entry.
11:30:25 AM	4624	Successful logon (two records)	10.209.222.48	Two records with an identical timestamp.
11:53:04 AM	4625	Failed logon (Audit Failure)	Not shown in list view	Observed in Figure 1.
11:53:06 AM	4625	Failed logon (Audit Failure)	Not shown in list view	Observed in Figure 1.
11:53:21 AM	4625	Failed logon (Audit Failure)	10.209.222.48 (per Figure 2 and task information)	Selected event in Figure 1; first row of the filtered list.
11:54:05 AM	4624	Successful logon	10.209.222.48	Observed 44 seconds after the 11:53:21 AM failed entry.
11:54:14 AM	4624	Successful logon	10.209.222.48	Observed 9 seconds after the previous record.
12:46:46 PM	4624	Successful logon	10.209.222.48	Observed.
1:18:28 PM	4624	Successful logon	10.209.222.48	Last successful logon record in the queried output.

The source IP of the failed entries other than the one attributed to Figure 2 is not visible in the Event Viewer list. Successful logons therefore appear close in time to failed entries, but the evidence does not show that those failed entries came from 10.209.222.48 or targeted the same account as the successful logons.

10. INCIDENT FINDINGS

10.1 Observations

- Event ID 4625 failed logon activity was observed in the Security log. The filtered view displayed 296 events out of 26,838 events in the log.

- The investigated source IP was 10.209.222.48.
- The target account shown in the selected 4625 event was Administrator, using logon type 3 and NTLM, with source port 58690.
- The PowerShell query returned a count of 43 for the failed-logon filtering on this source IP. A parsing error for one event is visible in the command output, so the value is not presented as a fully validated total.
- Event ID 4624 successful logons from the same source IP were observed, and the evidence shows 12 successful logon records.
- The evidence therefore demonstrates repeated authentication activity from the investigated source IP.

10.2 Interpretation

Evidence indicates repeated authentication activity from 10.209.222.48, consisting of failed network logons and successful logons within the same period. This pattern is suspicious and warrants further review. It is consistent with repeated authentication attempts, but the evidence does not establish a confirmed brute-force attack, and no final verdict is drawn.

10.3 Not demonstrated in the provided evidence

- The accounts and logon types of the 12 successful logon records.
- The source IP addresses of individual Event ID 4625 entries other than the inspected event.
- Whether any successful logon followed failed attempts against the same account.
- The identity, ownership, or role of the host at 10.209.222.48, and whether the successful logons were authorised.
- Any containment, escalation, or response action.

11. TRYHACKME PRACTICAL TRAINING

Both required TryHackMe rooms were completed. The screenshots serve as evidence of completion only; no task answers, flags, or room findings are shown or reported.

11.1 Investigating Windows

The room page (tryhackme.com/room/investigatingwindows) shows the description "A windows machine has been hacked, its your job to go investigate this windows machine and find clues to what the hacker might have done." The completion banner reads "Room completed: 100%", and Task 1 – Investigating Windows is marked complete.

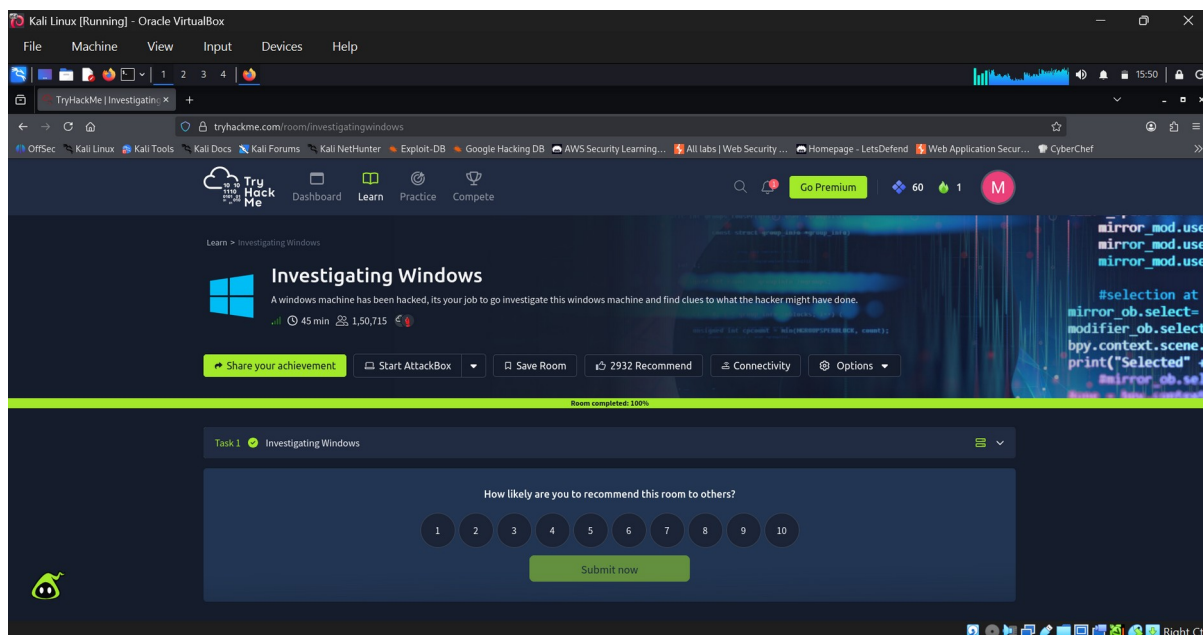


Figure 6 – 06_tryhackme_investigating_windows_completed.png
TryHackMe Investigating Windows room showing 100% completion.

11.2 Juicy Details

The room page (tryhackme.com/room/juicydetails) shows the description “A popular juice shop has been breached! Analyze the logs to see what had happened...”. The completion banner reads “Room completed: 100%”, and Task 1 – Introduction, Task 2 – Reconnaissance, and Task 3 – Stolen data are marked complete. Any task rows below these are not visible.

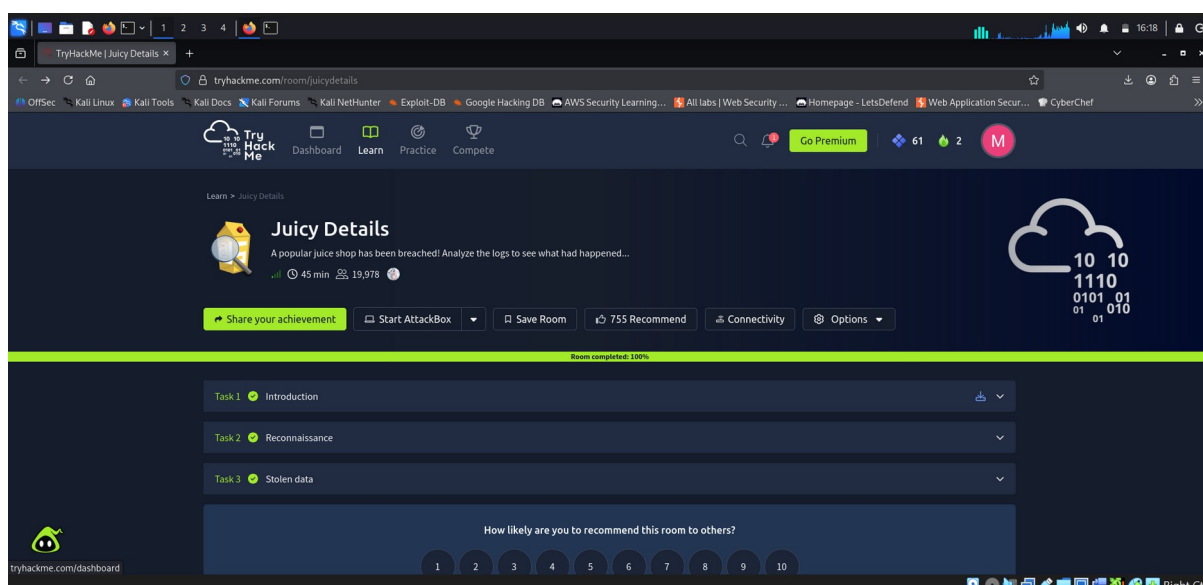


Figure 7 – 07_tryhackme_juicy_details_completed.png
TryHackMe Juicy Details room showing 100% completion.

12. SKILLS PRACTICED

Skill	Where demonstrated
Windows Security Log Analysis	Figures 1 and 2
Event ID 4625 Investigation	Figures 1, 2, and 3
Event ID 4624 Investigation	Figures 4 and 5

Skill	Where demonstrated
Authentication Event Analysis	Figure 2 (logon type, authentication package, status fields)
Source IP Identification	Figures 2, 3, 4, and 5
PowerShell-Based Log Filtering	Figures 3, 4, and 5
Incident Timeline Development	Section 9
Basic Incident Detection	Sections 5 to 10
TryHackMe Defensive Security Investigation Practice	Figures 6 and 7

13. SECURITY SIGNIFICANCE

Event ID 4625 records each failed logon and includes fields such as the target account, logon type, authentication package, and source IP address. Monitoring these events allows an analyst to see when failed authentication is concentrated on one account or comes from one source, which can be an early sign of credential guessing, although misconfigured services and users mistyping passwords also generate 4625 events.

Correlating Event ID 4625 with Event ID 4624 from the same source IP shows whether failed logons were accompanied or followed by successful ones. This affects triage priority, because a successful logon from a source that also produced failures needs closer review than failures alone. The correlation also depends on details that were not visible in this task, such as the account names on the 4624 records, and an analyst would need them before reaching a conclusion. Using PowerShell to filter the log by source IP makes this comparison repeatable, and building a timeline from the timestamps gives the sequence of events needed for an incident record.

14. CONCLUSION

This task provided practical experience in identifying and investigating authentication-related events in Windows Security logs. Event ID 4625 failures were filtered in Event Viewer, a source IP address (10.209.222.48) was identified, PowerShell was used to count failed logons for that address and to list 12 successful Event ID 4624 logons from it, and a timeline was built from the observed timestamps. The 43 count returned by the failed-logon query is reported with the XML parsing error that appeared in its output. The evidence indicates repeated authentication activity from the source IP but does not support a confirmed attack verdict. The required TryHackMe rooms, Investigating Windows and Juicy Details, were completed at 100%.

15. EVIDENCE / SCREENSHOT INDEX

Figure Number	Screenshot Filename	Description
Figure 1	01_security_log_event_4625_failed_logins.png	Event Viewer, Security log filtered for Event ID 4625 (296 events); selected event General tab (Section 5)
Figure 2	02_event_4625_source_ip_administrator.png	Event Properties Details tab: target account, source IP, logon type, NTLM, port (Section 6)
Figure 3	03_powershell_43_failed_login_attempts.png	PowerShell query for Event ID 4625 by source IP: Count 43, with visible XML parsing error (Section 7)
Figure 4	04_successful_logon_events_from_source_ip_10.209.222.48.png	PowerShell query listing 12 Event ID 4624 records matching 10.209.222.48 (Section 8)

Figure Number	Screenshot Filename	Description
Figure 5	05_failed_logon_attempt_count_from_source_ip_10.209.222.48.png	PowerShell XML-based listing of the same 12 Event ID 4624 records with SourceIP; file name refers to a failed logon count, content shows 4624 records (Section 8)
Figure 6	06_tryhackme_investigating_windows_completed.png	TryHackMe Investigating Windows, Room completed: 100% (Section 11.1)
Figure 7	07_tryhackme_juicy_details_completed.png	TryHackMe Juicy Details, Room completed: 100% (Section 11.2)